Chapter 11

How to create databases, tables, and indexes

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Objectives

- Learn how create, update, and drop databases.
- Learn how create, update, and drop tables.
- Learn how create, update, and drop indexes.

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1. How to work with databases.
2. How to work with tables.
3. How to work with indexes.

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1. How to work with databases

How to use the CREATE DATABASE statement

Syntax

```
CREATE DATABASE [IF NOT EXISTS] db_name
```

Attempt to create a database named AP

```
CREATE DATABASE ap
```

Create a database named AP only if it doesn't exist

```
CREATE DATABASE IF NOT EXISTS ap
```


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1. How to work with databases

How to use the DROP DATABASE statement

Syntax

```
DROP DATABASE [IF EXISTS] db_name
```

Attempt to drop a database named AP

```
DROP DATABASE ap
```

Drop a database named AP only if it exists

```
DROP DATABASE IF EXISTS ap
```


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1. How to work with databases

How to use the USE statement

Syntax

```
USE db_name
```

Select a database named AP

```
USE ap
```


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2. How to work with tables: Create table

The syntax of the CREATE TABLE statement

```
CREATE TABLE [db_name.]table_name
(
  column_name_1 data_type [column_attributes]
  [, column_name_2 data_type [column_attributes]]...
  [, table_level_constraints]
)
```


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2. How to work with tables: Create table

Common column attributes

Attribute	Description
`NOT NULL`	Indicates that the column doesn't accept null values. If omitted, the column can accept null values.
`UNIQUE`	Specifies that each value stored in the column must be unique.
`DEFAULT default_value`	Specifies a default value for the column as a literal or as an expression.
`AUTO_INCREMENT`	Identifies a column whose value is automatically incremented by MySQL when a new row is added. An auto increment column must be defined as an integer or a floating-point number.

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2. How to work with tables

A statement that creates a table without column attributes

```
CREATE TABLE vendors
(
  vendor_id       INT,
  vendor_name     VARCHAR(50)
)
```

A statement that creates a table with column attributes

```
CREATE TABLE vendors
(
  vendor_id       INT             NOT NULL     UNIQUE AUTO_INCREMENT,
  vendor_name     VARCHAR(50)     NOT NULL     UNIQUE
)
```


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2. How to work with tables: Create table

Another statement that creates a table with column attributes

```
CREATE TABLE invoices
(
  invoice_id        INT             NOT NULL     UNIQUE,
  vendor_id         INT             NOT NULL,
  invoice_number    VARCHAR(50)     NOT NULL,
  invoice_date      DATE,
  invoice_total     DECIMAL(9,2)    NOT NULL,
  payment_total     DECIMAL(9,2)                 DEFAULT 0
)
```


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2. How to work with tables: Primary key

The syntax of a table-level primary key constraint

```
[CONSTRAINT [constraint_name]]
PRIMARY KEY (column_name_1[, column_name_2]...)
```

A table with column-level constraints

```
CREATE TABLE vendors
(
  vendor_id       INT             PRIMARY KEY     AUTO_INCREMENT,
  vendor_name     VARCHAR(50)     NOT NULL        UNIQUE
)
```


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2. How to work with tables: Primary key

A table with table-level constraints

```
CREATE TABLE vendors
(
  vendor_id       INT             AUTO_INCREMENT,
  vendor_name     VARCHAR(50)     NOT NULL,
  CONSTRAINT vendors_pk PRIMARY KEY (vendor_id),
  CONSTRAINT vendor_name_uq UNIQUE (vendor_name)
)
```

A table with a two-column primary key constraint

```
CREATE TABLE invoice_line_items
(
  invoice_id                INT             NOT NULL,
  invoice_sequence          INT             NOT NULL,
  line_item_description     VARCHAR(100)    NOT NULL,
  CONSTRAINT line_items_pk PRIMARY KEY (invoice_id, invoice_sequence)
)
```


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2. How to work with tables: Primary key

A table with table-level constraints

```
CREATE TABLE vendors
(
  vendor_id       INT             AUTO_INCREMENT,
  vendor_name     VARCHAR(50)     NOT NULL,
  CONSTRAINT vendors_pk PRIMARY KEY (vendor_id),
  CONSTRAINT vendor_name_uq UNIQUE (vendor_name)
)
```

A table with a two-column primary key constraint

```
CREATE TABLE invoice_line_items
(
  invoice_id                INT             NOT NULL,
  invoice_sequence          INT             NOT NULL,
  line_item_description     VARCHAR(100)    NOT NULL,
  CONSTRAINT line_items_pk PRIMARY KEY (invoice_id, invoice_sequence)
)
```


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2. How to work with tables: foreign key

The syntax of a column-level foreign key constraint

```
[CONSTRAINT] REFERENCES table_name (column_name)
  [ON DELETE {CASCADE|SET NULL}]
```

The syntax of a table-level foreign key constraint

```
[CONSTRAINT constraint_name]
  FOREIGN KEY (column_name_1[, column_name_2]...)
  REFERENCES table_name (column_name_1[, column_name_2]...)
  [ON DELETE {CASCADE|SET NULL}]
```


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2. How to work with tables: foreign key

A table with a column-level foreign key constraint

```
CREATE TABLE invoices
(
  invoice_id        INT             PRIMARY KEY,
  vendor_id         INT             REFERENCES vendors (vendor_id),
  invoice_number    VARCHAR(50)     NOT NULL    UNIQUE
)
```

A table with a table-level foreign key constraint

```
CREATE TABLE invoices
(
  invoice_id        INT             PRIMARY KEY,
  vendor_id         INT             NOT NULL,
  invoice_number    VARCHAR(50)     NOT NULL        UNIQUE,
  CONSTRAINT invoices_fk_vendors
    FOREIGN KEY (vendor_id) REFERENCES vendors (vendor_id)
)
```


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2. How to work with tables: foreign key

An INSERT statement that fails because a related row doesn't exist

```
INSERT INTO invoices
VALUES (1, 1, '1')
```

The response from the system

```
Error Code: 1452. Cannot add or update a child row: a foreign key con-
straint fails ('ex'.'invoices', CONSTRAINT 'invoices_fk_vendors' FOREIGN
KEY ('vendor_id') REFERENCES 'vendors' ('vendor_id'))
```

A constraint that uses the ON DELETE clause

```
CONSTRAINT invoices_fk_vendors
  FOREIGN KEY (vendor_id) REFERENCES vendors (vendor_id)
  ON DELETE CASCADE
```


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2. How to work with tables: Alter table

The syntax for modifying the columns of a table

```
ALTER TABLE [db_name.]table_name
{
ADD            column_name data_type [column_attributes] |
DROP COLUMN    column_name |
MODIFY         column_name data_type [column_attributes] |
RENAME COLUMN old_column_name TO new_column_name
}
```


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2. How to work with tables: Alter table

A statement that adds a new column

```
ALTER TABLE vendors
ADD last_transaction_date DATE
```

A statement that drops a column

```
ALTER TABLE vendors
DROP COLUMN last_transaction_date
```


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2. How to work with tables: Alter table

A statement that changes the length of a column

```
ALTER TABLE vendors
MODIFY vendor_name VARCHAR(100) NOT NULL
```

A statement that changes the data type of a column

```
ALTER TABLE vendors
MODIFY vendor_name CHAR(100) NOT NULL
```

A statement that changes the default value of a column

```
ALTER TABLE vendors
MODIFY vendor_name VARCHAR(100) NOT NULL DEFAULT 'New Vendor'
```


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2. How to work with tables: Alter table

A statement that changes the name of a column

```
ALTER TABLE vendors
RENAME COLUMN vendor_name TO v_name
```

A statement that fails because it would cause data to be lost

```
ALTER TABLE vendors
MODIFY v_name VARCHAR(10) NOT NULL
```

The response from the system

```
Error Code: 1265. Data truncated for column 'v_name' at row 1
```


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2. How to work with tables: Alter constraints

The syntax for modifying the constraints of a table

```
ALTER TABLE [dbname.]table_name
{
ADD  PRIMARY KEY constraint_definition |
ADD  [CONSTRAINT constraint_name] FOREIGN KEY constraint_definition |
DROP PRIMARY KEY |
DROP FOREIGN KEY constraint_name
}
```

A statement that adds a primary key constraint

```
ALTER TABLE vendors
ADD PRIMARY KEY (vendor_id)
```


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2. How to work with tables: Alter constraints

A statement that adds a foreign key constraint

```
ALTER TABLE invoices
ADD CONSTRAINT invoices_fk_vendors
   FOREIGN KEY (vendor_id) REFERENCES vendors (vendor_id)
```

A statement that drops a primary key constraint

```
ALTER TABLE vendors
DROP PRIMARY KEY
```

A statement that drops a foreign key constraint

```
ALTER TABLE invoices
DROP FOREIGN KEY invoices_fk_vendors
```


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2. How to work with tables: rename, truncate, and drop a table

A statement that renames a table

```
RENAME TABLE vendors TO vendor
```

A statement that deletes all data from a table

```
TRUNCATE TABLE vendor
```

A statement that deletes a table from the current database

```
DROP TABLE vendor
```


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2. How to work with tables: rename, truncate, and drop a table

A statement that qualifies the table to be deleted

```
DROP TABLE ex.vendor
```

A statement that returns an error due to a foreign key reference

```
DROP TABLE vendors
```

The response from the system

```
Error Code: 3730. Cannot drop table 'vendors' referenced by a foreign key
constraint 'invoices_fk_vendors' on table 'invoices'
```


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3. How to work with indexes: create index

MySQL INDEX

Index Column

Pointers

Ordered List

P_8	Dan
P_6	David
P_3	Hannah
P_4	John
P_10	Jonas
P_11	Lily
P_2	Peter
P_9	Reema
P_1	Roger
P_7	Ryan
P_5	Sarah

	ID	First Name
P_1	1	Roger
P_2	2	Peter
P_3	3	Hannah
P_4	4	John
P_5	5	Sarah
P_6	6	David
P_7	7	Ryan
P_8	8	Dan
P_9	9	Reema
P_10	10	Jonas
P_11	11	Lily

Table name: Authors

*Query: SELECT * FROM authors where first_name = 'Jonas';*

3. How to work with indexes: create index

The syntax of the CREATE INDEX statement

```
CREATE [UNIQUE] INDEX index_name
  ON [dbname.]table_name (column_name_1 [ASC|DESC][,
                          column_name_2 [ASC|DESC]]...)
```

A statement that creates an index based on a single column

```
CREATE INDEX invoices_invoice_date_ix
  ON invoices (invoice_date)
```


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3. How to work with indexes: create index

A statement that creates an index based on two columns

```
CREATE INDEX invoices_vendor_id_invoice_number_ix
  ON invoices (vendor_id, invoice_number)
```

A statement that creates a unique index

```
CREATE UNIQUE INDEX vendors_vendor_phone_ix
  ON vendors (vendor_phone)
```


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3. How to work with indexes: create index

A statement that creates an index that's sorted in descending order

```
CREATE INDEX invoices_invoice_total_ix
  ON invoices (invoice_total DESC)
```

A statement that drops an index

```
DROP INDEX vendors_vendor_phone_ix ON vendors
```


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Thank you!

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